

```

function sub(x: int, y: int): int
  decreases(if (x <= 0) then -x else x) {
    if (x == 0) then -y
    else if (x > 0) then sub(x - 1, y - 1)
    else sub(x + 1, y + 1)
  }

// Computes y - x instead of x - y
function mySub(x: int, y: int): int
  decreases(if (y <= 0) then -y else y) {
    if (y == 0) then -x
    else if (y > 0) then mySub(x - 1, y - 1)
    else mySub(x + 1, y + 1)
  }

```

Listing 3.24: sub and mySub Functions

These two functions use the same variant to prove termination, and are equivalent if the arguments are matched correctly. Therefore, Dafny is able to prove their equivalence as follows, but only if the user explicitly copies the variant from one of the two functions:

```

lemma equivalenceSub(x: int, y: int)
  decreases(if (x <= 0) then -x else x)
  ensures (sub(x, y) == mySub(y, x))
{}

```

Listing 3.25: Equivalence Lemma for sub and mySub

In addition, if at least one of the functions requires a lemma or an assertion to prove termination, this must be copied into the equivalence lemma as well. For example, here are two implementations of a function to reduce a list down to its unique elements:

```

function solution_2(lst: List<int>): List<int>
  decreases(length(lst)) {
    match lst {
      case Nil => Nil
      case Cons(hd, tl) =>
        lemM(hd, tl);
        Cons(hd, solution_2(drop_2(tl, hd)))
    }
  }

function uniq2(lst: List<int>): List<int>
  decreases(length(lst)) {
    match lst {
      case Nil => Nil
      case Cons(hd, tl) =>
        lem2(hd, tl);
        Cons(hd, uniq2(drop(hd, tl)))
    }
  }

```

Listing 3.26: solution_2 and uniq2 Functions

The definitions of `drop_2` and `drop` are irrelevant for this example, so I will omit them. All that is required is that they are equivalent, when the arguments are matched correctly:

```

lemma equivalenceDrop(a: int, lst: List<int>)
  ensures (drop(a, lst) == drop_2(lst, a))
{}

```

Listing 3.27: Equivalence Lemma for drop_2 and drop

`solution_2` and `uniq2` also use helper lemmas `lemM` and `lem2` respectively to prove termination. These lemmas are equivalent, so I will only provide `lem2`:

```

lemma lem2(a: int, lst: List<int>)
  ensures length(drop(a, lst)) <= length(lst)
{}

```

Listing 3.28: lem2 Helper Lemma

Using `equivalenceDrop`, Dafny is able to prove the equivalence of `solution_2` and `uniq2`, but only if the user explicitly calls `lem2`:

```

lemma equivalence_solution2_uniq2(lst: List<int>)
  ensures (solution_2(lst) == uniq2(lst))
  decreases (length(lst))
{
  match lst
  case Nil => {}
  case Cons(hd, tl) =>
    lem2(hd, tl);
    equivalenceDrop(hd, tl);
}

```

Listing 3.29: Equivalence Lemma for `solution_2` and `uniq2`

Thus, if lemmas or assertions are needed to prove termination of at least one of the functions being compared, the user must explicitly copy them into the equivalence lemma as well. *repetition*

3.7 Equivalence of Imperative Methods

Recursive functions can be converted into their iterative versions. For example, the recursive implementation of `isSortedB`, described in Listing 3.2, can be converted into the following imperative method:

```

method isSortedBI(l: seq<int>) returns (res: bool)
  ensures res <==> (forall i :: 0 <= i < |l| - 1 ==> l[i] <= l[i+1])
{
  var currentL := 1;

  while true
    decreases |currentL|
    invariant currentL == 1[|l| - |currentL| .. ]
    invariant forall i :: 0 <= i < |l| - |currentL| ==>
      (i < |l| - 1 ==> l[i] <= l[i+1])
  {
    if |currentL| == 0 {
      return true;
    } else if |currentL| == 1 {
      currentL := [];
    } else {
      var x := currentL[0];
      var y := currentL[1];
      if x > y {
        return false;
      }
      currentL := currentL[1..];
    }
  }
}

```

Listing 3.30: `isSortedBI` Method

Proving the equivalence of imperative methods cannot be achieved directly through lemmas in Dafny; only functions can be called within lemmas as, by definition, Dafny functions are pure while Dafny methods are allowed to have side effects. Therefore, to prove equivalence, we must prove postconditions for both methods, and then state that these postconditions are equivalent. In turn, as in the above example, the user will often need to provide loop invariants and variants to help Dafny prove these postconditions.

These specifications can be very difficult to write correctly. Additionally, just proving that the methods' postconditions are equivalent is not sufficient to prove the equivalence of the methods themselves, as the postconditions may not fully capture the behaviour of the methods. *true*

3.7.1 Proving Equivalence of Imperative Methods using Product Programs

As described by Lahiri et al. [8], one solution to these difficulties is to generate product programs. A product program is a method that contains the bodies of the two methods being compared, in such a way that relationships between the two can be verified.

For example, we can use this process to prove the equivalence of the imperative versions of `isSortedR` and `isSortedB` (the function versions can be found in Listings 3.1 and 3.2 respectively). First, the imperative version of `isSortedR` is as follows, requiring a helper function `loopI`:

```

method loopI(p: int, l: seq<int>) returns (res: bool)
  ensures res <==> ((forall i :: 0 <= i < |l| - 1 ==> l[i] <= l[i+1]) &&
    (|l| > 0 ==> p <= l[0]))
{
  var currentP := p;
  var currentL := 1;

  while true
    decreases |currentL|
    invariant currentL == 1[|l| - |currentL| .. ]
    invariant currentP == if |l| == |currentL| then p
                          else 1[|l| - |currentL| - 1]
    invariant |l| > |currentL| ==> p <= l[0]
    invariant forall i :: 0 <= i < |l| - |currentL| - 1 ==> l[i] <= l[i+1]
  {
    if |currentL| == 0 {
      return true;
    } else {
      var x := currentL[0];
      if currentP > x {
        return false;
      }
      currentP := x;
      currentL := currentL[1..];
    }
  }
}

method isSortedRI(l: seq<int>) returns (res: bool)
  ensures res <==> (forall i :: 0 <= i < |l| - 1 ==> l[i] <= l[i+1])
{
  if |l| == 0 {
    return true;
  } else {
    var x := l[0];
    var xs := l[1..];
    var result := loopI(x, xs);
    return result;
  }
}

```

Listing 3.31: isSortedRI Method with loopI Helper Method

These methods can be used to construct a product program as follows, using the definition of loopI given above:

```

method isSortedBI_RI_Equivalence(
  lb: seq<int>,
  lr: seq<int>
) returns (rres: bool, bres: bool)
  requires lb == lr
  ensures rres == bres
{
  // Defining output variables
  var rout: bool;
  var bout: bool;

  // Computing isSortedRI
  if |lr| == 0 {
    rout := true;
  } else {
    var x := lr[0];
    var xs := lr[1..];
    var result := loopI(x, xs);
    rout := result;
  }

  // Computing isSortedBI
  var currentL := lb;

  while true
    decreases |currentL|
    invariant currentL == lb[|lb| - |currentL| .. ]
    invariant forall i :: 0 <= i < |lb| - |currentL| ==>

```

```

                                (i < |lb| - 1 ==> lb[i] <= lb[i+1])
{
  if |currentL| == 0 {
    bout := true;
    break;
  } else if |currentL| == 1 {
    currentL := [];
  } else {
    var x := currentL[0];
    var y := currentL[1];
    if x > y {
      bout := false;
      break;
    }
    currentL := currentL[1..];
  }
}

// Returning the results
return rout, bout;
}

```

Listing 3.32: Product Program for `isSortedBI` and `isSortedRI` using `loopI`

interesting idea
(new to me)

The specification of this product program ensures that, since Dafny is able to verify this method, `isSortedBI` and `isSortedRI` must be equivalent. Therefore, this process removes the need for the user to provide postconditions for the individual methods, that capture their behaviour fully. In this case, a postcondition for the `loopI` helper function is still required, but the body of this method can also be copied into the product program, removing this requirement as well. However, the user must still provide the loop invariants and variants for the individual methods.

can you auto-gen these product programs (maybe without the invariants)?

3.7.2 Equivalence of Methods and Functions

A similar problem to that discussed above is proving the equivalence of a method and a function. In this case, the user does not need to write a postcondition for the method, since the function can be called directly within the specification. The user still needs to provide loop invariants and variants for the method, however, these do not need to be as complicated as those detailed above, as again, they can be defined in terms of the function. For example, the function `isSortedB` defined in Listing 3.2 can be proven to be equivalent with the following imperative method:

```

method isSortedBI(l: seq<int>) returns (res: bool)
  ensures res == isSortedB(l)
{
  var currentL := l;

  while true
    decreases |currentL|
    invariant isSortedB(currentL) == isSortedB(l)
  {
    if |currentL| == 0 {
      return true;
    } else if |currentL| == 1 {
      currentL := [];
    } else {
      var x := currentL[0];
      var y := currentL[1];
      if x > y {
        return false;
      }
      currentL := currentL[1..];
    }
  }
}

```

yes
much nicer/cleaner

Listing 3.33: `isSortedBI` Method with simplified specifications

Thus, the necessary specifications can be greatly simplified if they are defined in terms of the function the method is being compared to.

3.8 Equivalence of Higher-Order Functions

Dafny is able to prove the equivalence of some higher-order functions, but not all. For example, here are two equivalent functions `andThen1` and `andThen2`:

```
function andThen1<A, B, C>(f: A -> B, g: B -> C): A -> C { a => g(f(a)) }
function andThen2<A, B, C>(ff: A -> B, gg: B -> C): A -> C { aa => gg(ff(aa)) }
```

behavior?

Listing 3.34: `andThen1` and `andThen2` Functions

Dafny is able to prove their equivalence without any help:

```
lemma equivalenceAndThen<A, B, C>(f: A -> B, g: B -> C)
  ensures andThen1(f, g) == andThen2(f, g)
{}
```

Listing 3.35: Equivalence Lemma for `andThen1` and `andThen2`

Here are two more such functions `flip1` and `flip2`:

```
function flip1<A, B, C>(f: (A, B) -> C): (B, A) -> C {
  (b, a) => f(a, b)
}

function flip2<A, B, C>(f: (A, B) -> C): (B, A) -> C {
  (b, a) => var res := f(a, b); res
}
```

behavior?

Listing 3.36: `flip1` and `flip2` Functions

Dafny is able to prove that the outputted functions give the same output for the same inputs:

```
lemma equivalenceFlip<A, B, C>(f: (A, B) -> C)
  ensures forall b: B, a: A :: flip1(f)(b, a) == flip2(f)(b, a)
{}
```

Listing 3.37: Equivalence Lemma for `flip1` and `flip2` with specified arguments

However, it is unable to prove that the outputted functions themselves are equivalent:

```
// Dafny cannot prove this lemma
lemma equivalenceFlip<A, B, C>(f: (A, B) -> C)
  ensures flip1(f) == flip2(f)
{}
```

Listing 3.38: Invalid Equivalence Lemma for `flip1` and `flip2`

Therefore, Dafny is able to prove the equivalence of some higher-order functions. However, it struggles with others, despite being able to prove that the outputted functions give the same output for the same inputs.

any ideas where the limitations truly lie
(i.e. any pattern, to what it can/cannot prove)?

Chapter 4

Plans

4.1 Project Plan

The aim of this project is to build a tool on top of the Dafny program verifier that removes as many of the limitations identified in Chapter 3 as possible. Some of these limitations are out of the scope of this project. Preventing incorrect counterexamples (as described in Section 3.2) is one such limitation, as it results from problems in Boogie and the underlying Z3 solver. Similarly, the limitations related to imperative programs (as described in Section 3.7) are also out of scope, as Dafny functions provide enough complexity for this project by themselves.

I plan to build my tool in Scala. It will be called from the command line, taking as input multiple Dafny files, containing the functions to be compared and any help the user needs to provide. The tool will also require a JSON configuration file, specifying which functions must be proven to be equivalent and any other information the tool requires. The tool will attempt to create an equivalence proof and prove it using Dafny; Dafny's output will be used as the output of the tool.

The following table summarises the current development plan, broken down into milestones. Each milestone has an associated completion deadline and a description of what that milestone involves. Each milestone is described as either Necessary or Optional. If Necessary milestones take longer than I expected, I will use the time previously dedicated to Optional milestones to complete the Necessary ones.

If I have time remaining once all the project and evaluation milestones are completed, I plan to further extend our tool to handle the remaining limitations relating to Dafny functions identified in Chapter 3. This includes removing the necessity for the user to explicitly define induction schemes to prove lemmas (Section 3.5) and implementing the ability to prove user-provided auxiliary lemmas (Section 3.4).

Milestone	Description	Deadline	Necessity
1	Building the basic structure	09/02/2026	Necessary
2	Creating a CI script for Continuous Testing and Evaluation	16/02/2026	Necessary
3	Create Parser for Dafny functions	02/03/2026	Necessary
4	Copying Variants into the Main Lemma	09/03/2026	Optional
5	Generating Helper Equivalence Lemmas as Axioms	16/03/2026	Optional
6	Calling Helper Lemmas in the Main Lemma	23/03/2026	Optional
7	Calling Helper Lemmas in Equivalence Lemmas	26/03/2026	Optional
8	Proving Equivalence with one of a number of solutions	02/04/2026	Optional
9	Allowing the user to provide Auxiliary Lemmas as Axioms	06/04/2026	Optional
10	Proving equivalence of higher order functions	01/05/2026	Optional
11	Writing the Final Report	01/06/2026	Necessary

Table 4.1: Summary of the current development plan

Building the basic structure This milestone involves creating the skeleton of the tool. It should be able to take in Dafny files and a JSON configuration file. The Dafny files will contain the two functions being compared and an equivalence lemma without any user assistance. The tool will simply combine these files together, call the Dafny verifier on the resulting file, and return its output. Equivalence

lemmas with no user assistance are used here to provide an evaluation baseline for the evaluation of the remaining milestones.

Creating a CI script for Continuous Testing and Evaluation At each milestone, evaluation data must be generated and collected (see Section 4.2 for more details on this). This milestone involves creating a script to collect this data, and setting up a GitLab CI pipeline to automatically run this script on every merge. Once this is done, evaluation data should be collected for the previous milestone. Then, the equivalence lemmas should be replaced with lemmas that include all the user assistance that Dafny requires, followed by a recollection of the evaluation data, to serve as another baseline for the remaining milestones. ✓

Create Parser for Dafny functions Several of the future milestones will require an in depth understanding of the structure of the Dafny functions being compared and of the equivalence lemma provided by the user. Therefore, in this milestone, I will create parsers specifically for Dafny functions and lemmas. ✓

Copying Variants into the Main Lemma As described in Section 3.6, when one or both of the functions being compared require the user to explicitly provide variants, this must be copied into the main equivalence lemma. This milestone involves implementing this functionality. ✓

core | **Generating Helper Equivalence Lemmas as Axioms** This milestone will involve implementing the method of finding and generating equivalence lemmas for helper functions described by Milovančević and Kunčak [11] and summarised in Section 3.3. For now, these lemmas will be generated as axioms; they will be proven properly in a later milestone. ✓

core | **Calling Helper Lemmas in the Main Lemma** As described in Sections 3.3 and 3.6, helper equivalence and termination lemmas must be explicitly called in the main equivalence lemma, under the correct conditions and with the correct arguments. This milestone involves implementing this specifically for the main equivalence lemma. ✓

core | **Calling Helper Lemmas in Equivalence Lemmas** This milestone involves extending the functionality implemented in the previous milestone to also prove the generated helper equivalence lemmas. ✓

Proving Equivalence with one of a number of solutions In certain cases, including those described by Milovančević and Kunčak [11], the user may wish to prove the equivalence of a function with any one of a number of possible other functions. This milestone involves implementing this. ✓

Allowing the user to provide Auxiliary Lemmas as Axioms In certain cases, the user will need to provide auxiliary lemmas to help Dafny prove equivalence. These lemmas are specific to each individual scenario, and so automatic generation is out of scope for this project. Therefore, a reasonable alternative is to allow the user to provide auxiliary lemmas, which can then be called in the main equivalence lemma. These auxiliary lemmas will be added as axioms, as it would be difficult to prove them automatically. The burden of ensuring these lemmas are correct will be on the user. This milestone will involve implementing this. *are these not just added to the Dafny code input - seems trivial to support...*

Proving equivalence of higher order functions When proving the equivalence of functions that output lambda functions, these functions are equivalent if the lambda functions output the same values for the same inputs. This can be proven through a universal quantification over the domain of the lambda functions. This milestone involves implementing this functionality. ✓

Writing the Final Report Throughout development, I will add to the Final Report as I go along; this milestone involves completing the technical chapters in it. ✓

4.2 Evaluation Plan

In a similar manner to Section 4.1, here is a table summarising the current evaluation plan, broken down into milestones.

Milestone	Description	Deadline	Necessity
1	Creating Initial Evaluation Testsuite	Completed	Necessary
2	Setup of Evaluation Data Gathering	16/02/2026	Necessary
3	Testsuite Development	08/05/2026	Optional
4	Benchmark Comparison	11/05/2026	Necessary
5	Writing the Final Report	05/06/2026	Necessary

Table 4.2: Summary of the current evaluation plan

this is not part of eval plan

Creating Initial Evaluation Testsuite Before evaluation can begin, we need a testsuite containing pairs of functions which our tool will attempt to prove equivalent. As described in Chapter 3, I have already created an initial testsuite using the equivalence tests from the repository for the Stainless Scala verification tool[14]. This testsuite will be extended in a later milestone. *nice to have a precise testset*

Setup of Evaluation Data Gathering As described in Section 4.1, once each milestone is completed, a CI script will be used to automatically collect evaluation data. The first two milestones will provide baseline data on Dafny without any user assistance, and on Dafny with all the assistance it requires respectively. Data collected from all subsequent milestones will be compared to this baseline data. The evaluation data will consist of two categories. First, we must gather data to represent how difficult each example is for Dafny to verify. This category will include the metrics of the number of lines of code in each function being compared, the number of helper functions called and the time taken for Dafny to verify the example. Second, we must gather data to represent how much user assistance Dafny requires. This will include the metrics of the number of lines of code in the main equivalence lemma and the numbers of helper equivalence and auxiliary lemmas the user needs to provide. The aim of this project is for the metrics in the second category to decrease as much as possible. The first category metrics will ensure that we are interpreting these decreases in the correct context. These metrics are too shallow to provide a complete picture of the complexity of the examples and the level of user assistance required, however, they will provide a reasonable approximation.

Testsuite Development A second set of program equivalence tests have been gathered by Badihi, Li, and Rubin [1]. This milestone involves converting these tests from C and Java into Dafny, and appending them into my evaluation testsuite. *good idea*

Benchmark Comparison In this milestone, I will compare the results of our tool on our evaluation testsuite with the results from the original Dafny verifier, to demonstrate the improvements made. I will also compare our tool with Stainless on the subset of our testsuite that originated from their equivalence tests, to demonstrate how our tool compares with other state-of-the-art program verifiers.

Writing the Final Report Throughout development, I will add to the Final Report as I go along; this milestone involves completing the evaluation chapters in it.

not eval

how measure "level of user assistance"

different kind of user annotations have different difficulty levels to the user

+ some system profiling

Chapter 5

Ethical, Legal, Professional and Societal Issues

This project raises very few ethical, legal, professional and societal issues. The only potential issue is that our tool must obey the licenses of the any third-party libraries or tools it uses. In particular, our tool must obey Dafny's MIT license, since Dafny will be used to verify the final equivalence lemmas. It must also obey Stainless's Apache 2.0 license[14] and EqBench's MIT license[1], since these repositories ✓ will be used to form our evaluation testsuite.

what about environmental impact?

how sustainable is your tool - what are the power consumption overheads?

soundness? ← responsibility to not
say program is correct if
they can not
